SDC HACKATHON 2026

Technical & Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 1st November 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

1. Team & Project Overview

Team Name	Yaponomics
Project / Solution Name	Yapoz
Name (Leader)	Abdulrahman Imaad Mohamed
Email & Phone (Leader)	swe2409099@xmu.edu.my, +60173793811
University Affiliation(s)	Xiamen University Malaysia
Live Prototype URL	https://yapoz.vercel.app (judges: use the demo buttons on the login page)
Code Repository URL (public / access granted to judges)	https://github.com/Ramilbey/yapoz

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

Customers write to a business on WhatsApp or Telegram, or through the website simulator. The channel sends a signed webhook to a Next.js app on Vercel, which checks the signature and saves the message as a job in PostgreSQL on Supabase. Failed jobs retry. A pipeline loads the business's catalogue, hours, policies and FAQs, then a tool-calling AI agent replies. An AI router tries the KrackedDevs gateway first, then Groq (gpt-oss), then Google Gemini. The agent can only state facts that tools read from the database. Code prices the order, checks stock and saves it as pending. Receipt photos are read by a vision model and compared with the order by rules. The owner or a service user approves or rejects on the dashboard (Next.js, React). The backend checks role, status, stock and receipt before it acts. The customer is told in the same chat, stock is updated, and an event is written to the activity log. Live updates use PostgreSQL LISTEN/NOTIFY. Cloudflare Turnstile and Resend protect sign-up. Four Orbyte events are built and signed but not sent: there is no live Orbyte connection.

2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 7.0.2	Language for the whole codebase	Apache-2.0
Next.js 16.4.2	Web app, API routes, server rendering	MIT
React and React DOM 19.3.1	User interface	MIT
Tailwind CSS 4.3.3	Styling	MIT

Name & Version	Purpose in Project	License
Radix UI: dialog 1.3.1, dropdown-menu 2.2.1, popover 1.3.1, switch 1.4.1, tabs 1.2.1, tooltip 1.4.1	Accessible interface components	MIT
SWR 2.5.1	Client data fetching and live refresh	MIT
Drizzle ORM 0.45.4, Drizzle Kit 0.31.11	Database access and migrations	Apache-2.0, MIT
postgres (postgres.js) 3.4.9	PostgreSQL driver	Unlicense
Zod 4.6.7	Validation of every API request	MIT
jose 6.2.13, bcryptjs 3.0.3	Signed sessions, password hashing	MIT, BSD-3-Clause
grammY 1.46.0	Telegram bot framework	MIT
OpenAI SDK 7.39.0	Client for OpenAI-compatible AI APIs (KrackedDevs gateway, Groq, Gemini)	Apache-2.0
Anthropic SDK 0.135.0	Optional Claude adapter, not enabled in production	Apache-2.0
web-push 3.6.7	Browser push alerts to staff	MPL-2.0
unpdf 1.9.1, pdf-lib 1.17.1, qrcode 1.5.4	Read PDFs, generate sample PDFs, QR codes for customer links	MIT
date-fns 4.4.0, @date-fns/tz 1.5.0, city-timezones 1.3.4	Dates, time zones and place lookup	MIT
clsx 2.1.1, server-only 0.0.1	Class names, server-only module guard	MIT
Three.js 0.186.1, React Three Fiber 9.8.2, drei 10.7.9, postprocessing 6.39.5	3D landing page	MIT (postprocessing: Zlib)
GSAP 3.15.0, @gsap/react 2.1.2, Motion 14.4.0, Lenis 1.3.27	Animation and smooth scrolling	GSAP Standard "no charge" licence; MIT
Fontsource: Bricolage Grotesque, JetBrains Mono, Instrument Serif	Web fonts	OFL-1.1

Name & Version	Purpose in Project	License
tsx 4.23.15, ESLint 10.13.0, Playwright 1.64.1, fflate 0.8.3	Development tooling: scripts, linting, screenshots	MIT, Apache-2.0

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
KrackedDevs LiteLLM gateway (litellm.rapidscreen.io)	AI replies, message classification and tool calling. First choice when its key is set; the other providers are the fallback	sponsor-provided
Groq API (Groq)	AI replies, classification and tool calling (gpt-oss models; Llama and Qwen as fallback)	free
Gemini API (Google)	Reading receipt and menu photos (vision); chat fallback	free
WhatsApp Cloud API (Meta)	Receive customer messages, send replies, download receipt files (Meta test number)	free
Telegram Bot API (Telegram)	Receive customer messages, send replies, download receipt files	free
Google Calendar API (Google)	One-way copy of approved bookings to a business's calendar; off unless the business connects it	free
Cloudflare Turnstile (Cloudflare)	Human check at sign-up	free
Resend (Resend)	Sends the 6-digit sign-up code by email	free
Web Push (browser push services)	Alerts to staff browsers that have turned them on	free
Anthropic, OpenAI, OpenRouter and Ollama APIs	Supported by the AI router but disabled in production: no keys are set	not used

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Vercel (Vercel Inc.)	Hosts the web app and API as serverless functions and runs the daily retry sweep. Deployed from the GitHub main branch	Hobby (free), functions in Washington D.C. (iad1)
Supabase (PostgreSQL)	Database, job queue, live dashboard updates	Free tier, Seoul (aws-0-ap-northeast-2)
GitHub	Public source code hosting; pushes to main trigger the Vercel deployment	Free
Cloudflare (Turnstile)	Sign-up human check	Free
Resend	Email delivery for sign-up codes	Free
Replit	Used during development for the initial project structure; the app is not deployed there	Sponsor-provided, not a deployment region

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
gpt-oss-120b and gpt-oss-20b	OpenAI, served by Groq	pre-trained, used as is	Apache-2.0
Llama and Qwen models (fallback only; the version is chosen automatically from Groq's list)	Meta and the Qwen team (Alibaba), served by Groq	pre-trained, used as is	As published on Groq's model pages
Gemini Flash and Flash-Lite (newest stable version, chosen automatically)	Google, Gemini API	pre-trained, used as is	Google Gemini API terms
[Model id from the team's KrackedDevs key card]	Model vendors, served through the KrackedDevs gateway	pre-trained, used as is	Gateway and vendor terms; retention and training terms not stated to us

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Forno Sette (FORNO7), Kasih Hair Studio (KASIH1) and Dapur Mak Long Catering (CATER1): menus, prices, hours, FAQs, bookings, orders and demo conversations	Created by the team (seed scripts and data in the repository)	synthetic	Created by the team (Apache-2.0 with the repository)	N

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Synthetic orders and sample receipts for the 20-order measurement run	Generated by the team's script (scripts/synthetic-orders.ts)	synthetic	Created by the team	N
Test conversations and test sign-ups on the live deployment	Messages sent by team members and testers	private, self-collected	Not shared	Y (display names, chat IDs, a WhatsApp number, email addresses)

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☒ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Claude Code (Anthropic), with Claude Sonnet 5.5 and Opus 5.5	Writing and testing code and database migrations, debugging, and drafting the hackathon documents	major
Replit Agent	Initial project structure	moderate
OpenAI Codex	Coding, development and debugging assistance	moderate
ChatGPT (OpenAI)	Pitch script, presentation planning, documentation and README assistance	minor
Claude (Anthropic)	Slide-deck generation and presentation assistance	minor

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

AI-written code is kept only after the TypeScript check, the production build and the automated test suite pass (over 250 tests, run on a throwaway database). The team tests changes on the live app by chatting as a customer, and reviews each change before it is committed. Prices, stock and payment details shown to customers come from the owner's data, never from model memory. The approval gate and activity log are enforced in backend code and covered by tests. The team is responsible for the code, licences and security, and can explain any part of it.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☒ Yes ☐ No

If yes, list them below:

Sponsor	**Resource (API / credits / platform / data)**	**How It Was Used in the Project**
KrackedDevs (via the SDC Hackathon)	AI API credit, US$100 per team, as personal OpenAI-compatible keys	The AI router calls the KrackedDevs gateway first for replies and tool calling. Groq and Gemini answer if it fails or the credit ends (7 Nov 2026)
Replit (via the SDC Hackathon)	Replit Core plan and Agent credits	Replit Agent produced the initial project structure. The credits ran out, so development continued with Claude Code
Runpod (via the SDC Hackathon)	Runpod credit, US$25 per team	Not used: the team neither trains nor hosts a model

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

No. The AI router tries the KrackedDevs gateway first and falls back to Groq and Google Gemini on any failure, so Yapoz works without it. Groq and Gemini were the main providers before the gateway was added. Its credit ends on 7 Nov 2026, after which calls fail and the other providers answer. Replit produced only the initial project structure. Runpod credits were not used because the team neither trains nor hosts a model.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

Yapoz stores what customers send to a business: display name or username, chat ID or WhatsApp number, message text, photos and PDFs (including payment receipts), and the orders and bookings made from them. Staff accounts store a name, email and password hash. Message text goes to the AI providers in use (KrackedDevs gateway, Groq, Google Gemini); receipt images go only to vision-capable ones. The gateway forwards to model vendors, and we do not know its retention or training terms. We do not train models on this data. Channel credentials are encrypted (AES-256-GCM), passwords use bcrypt, every record belongs to one workspace, and secrets are redacted from logs. A business can set a 30 to 365 day purge, and finished orders past it are anonymised. The default is to keep data. All business data is synthetic.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Hallucination: the assistant can only state facts returned by tools that read the owner's data, prices are computed by code, and unanswered questions go to the owner. Receipt reading can misread or be fooled by an edited image, so its output is advice, the owner approves every order, and text on a receipt is treated as data, not instructions (covered by injection tests). Provider dependency: the gateway credit ends on 7 Nov 2026, and Groq and Gemini free tiers have rate limits and change models, so the router falls back between providers and retries jobs. If all fail, the customer gets a holding message and staff get an alert. WhatsApp replies are only possible within 24 hours of a customer message. No real business owner has tested Yapoz yet.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☒ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☒ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☒ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☒ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☒ Any use of official SDC sponsor resources has been declared in Section 5.
- ☒ The prototype described here is accessible to judges through the links provided in Section 1.
- ☐ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]

[Date]